UNIVERSITY OF MAKATI

College of Computing and Information Sciences

PROJECT CHARTER

Group Number:	**Date:**	**Revision Number:**
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Project Name: Development of QuizArena: A Web-Based Gamified Multiplayer Educational Online Quiz Battle System Using Retrieval-Augmented Generation (RAG), Adaptive Randomization, and LLM Integration		
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I. Project Summary and Goals

Contemporary higher education, particularly in logic-heavy disciplines such as Mathematics and Computer Science, has largely been served by gamified quiz platforms that remain limited to speed-based factual recall and rigid multiple-choice formats. These existing tools lack the specialized depth required to evaluate algorithmic logic, computational thinking, or multi-step problem-solving, and current educational technologies tend to operate in strict computational silos that leave a clear gap in dynamic, reasoning-based assessment.

Existing platforms further show a consistent lack of tailored AI solutions for analyzing complex, open-ended student input within collaborative settings, as scaling and consistently capturing the nuance of multi-step responses remains a significant challenge compared to simple multiple-choice grading. Poorly structured competitive environments have also been shown to foster toxic behavioral dynamics and to exacerbate performance disparities by disproportionately rewarding speed over deliberate reasoning. Combined with systemic educator concerns about heavy grading workloads and academic dishonesty, these limitations point to a substantial need for a dedicated, AI-driven assessment model capable of reliably evaluating learning methodologies and performance.

To address these challenges, this research proposes the development of QuizArena, a web-based, gamified multiplayer educational quiz battle system. QuizArena employs Retrieval-Augmented Generation (RAG) pipelines, an Adaptive Randomization algorithm, and an LLM-powered Solution Analyzer to precisely track and interpret the dynamic cognitive processes exhibited by learners in Mathematics, Computer Science, and English. The system automatically generates high-quality, contextually accurate assessment items directly from instructor-provided syllabi via the RAG architecture, with every generated question displaying its retrieved syllabus source for professor verification, paired with editing interfaces that allow educators to manually validate AI-generated content. Core features include flexible multiplayer modes, an LLM-driven Solution Analyzer capable of evaluating handwritten mathematical inputs and programming logic step-by-step, classification of student errors into misconception categories with class-level misconception analytics, and an evaluator that assesses programming logic rather than only final outputs.

The general objective of the study is to design and develop QuizArena to facilitate balanced academic competition and collaborative reasoning across English, Mathematics, and Coding from course syllabi, handwritten sketches, and programming inputs using RAG pipelines, Adaptive Randomization, and LLM integrations. In doing so, the platform is intended to automate academic assessments, mitigate speed-induced cognitive anxiety, and enhance collaborative learning, making the educational experience more convenient and engaging for both students and instructors.

Specific Objectives

1. Examine and understand the fundamental pedagogical mechanics of flexible multiplayer gamified learning, team-based bracketing, and the collaborative reasoning methodologies used in assessing diverse academic subjects like English, Mathematics, and Coding.
2. Explore existing gamified educational platforms and artificial intelligence frameworks relevant to the development of automated retrieval-augmented question generation, visual sketch analysis, and automated code evaluation systems.
3. Analyze, understand, and perform preprocessing activities on unstructured instructional materials, curriculum syllabi, and student-submitted mathematical and programming data gathered toward the development of the RAG pipeline and automated grading algorithms.
4. Train and evaluate the RAG pipeline, Adaptive Randomization algorithms, and LLM-driven Solution Analyzers using the Retrieval-Augmented Generation Assessment Scale (RAGAS) for Contextual Precision and Answer Relevancy, and Demographic Parity and Bounded Individual Loss for competitive matchmaking and team fairness calibration.
5. Design and develop a web-based application with the key features detailed under Project Scope below, including misconception-type classification within the Solution Analyzer and syllabus source citation for AI-generated questions.
6. Test the functional requirements (AI question generation accuracy, solution analyzer accuracy, multiplayer state synchronization, and teacher editing tools) and non-functional requirements (performance efficiency, throughput, load testing, and stress testing) of the system based on defined test cases.
7. Evaluate the system using metrics based on the ISO/IEC 25010 standard, focusing on Functional Suitability, Performance Efficiency, Usability, and Reliability.
8. Implement and deploy the system.

This project presents an opportunity to bridge the gap between interactive gamified engagement and rigorous algorithmic assessment. QuizArena is designed specifically for the Philippine higher-education context, positioning it to directly close research gaps identified in existing platforms: the absence of tools that assess deep, multi-step cognitive reasoning in logic-heavy subjects, the lack of AI-generated content grounded in validated curricula, the absence of fairness-calibrated matchmaking, and the lack of purpose-built solutions for Philippine higher education.

II. Project Scope (System Features)

The scope of the project centers on the application of Retrieval-Augmented Generation (RAG), Adaptive Randomization, and an LLM Solution Analyzer, specifically adapted for parsing instructional materials to autonomously generate questions, evaluating handwritten math sketches or programming code, and dynamically calibrating team brackets. These algorithms form the backbone of the application's automated assessment and equitable matchmaking capabilities.

System Components (User Roles)

- Professor / Educator — uploads and validates syllabi and instructional materials, edits and refines AI-generated questions, and accesses the Professor Dashboard for real-time match analytics.
- Student — participates in multiplayer quiz battles across individual, team, and Battle Royale formats, and submits handwritten mathematical solutions or programming code for evaluation.

Note: The research does not define a persona/module breakdown (e.g., SuperAdmin/Admin/User) with the same granularity as in other University of Makati capstone documentation; the two roles above are drawn from the Professor and Student user types referenced in the Chapter III system design artifacts (Use Case Diagram).

Key System Features

- Automates the generation of lesson-aligned assessment items using a professor-validated RAG pipeline.
- Displays the retrieved syllabus source (document, topic/section, and page) for every AI-generated question, allowing professors to verify content origin and improving the transparency and trust of AI-generated material.
- Provides manual editing tools for educators to validate, modify, and refine AI-generated questions.
- Features an LLM-powered Solution Analyzer to evaluate handwritten or sketched mathematical step-by-step solutions and programming logic.
- Classifies student errors identified by the Solution Analyzer into misconception categories (e.g., sign error, off-by-one error, syntax error, logical error, computational mistake) and aggregates them into class-level misconception analytics for professors.
- Features a real-time, flexible multiplayer quiz system synchronized via WebSockets, supporting individual seatwork, small teams, and Battle Royale formats.
- Allows dual scoring modes (Speed Mode and Discussion Mode) with a designated leader tie-breaking mechanic to resolve voting stalemates in collaborative play.
- Seeding and bracketing system utilizing the Adaptive Randomization algorithm for skill-based team matchmaking.
- Utilizes a comprehensive Professor Dashboard to visualize real-time match analytics, class-level misconception insights, and sortable results (e.g., by accuracy or alphabetically).
- Manages section registration, student join requests, and section reassignment of students per teacher for student filtering.
- Provides an adaptable, responsive view of the web application on both desktop and mobile devices.

Geographical and Temporal Scope

The geographical scope of this research extends to the University of Makati, targeting the student and faculty population within the College of Computing and Information Sciences. Data collection, testing, and user interaction are expected to occur within this defined geographical region. The study encompasses data collected and system development activities conducted during the period from May 2026 to May 2027; technological advancements beyond this temporal scope are not considered in the study.

Limitations

- The effectiveness of the application is contingent on the performance and accuracy of the RAG and LLM algorithms; constraints such as retrieval hallucinations or highly illegible handwritten sketches may impact performance and mandate human oversight via professor editing tools.

- Findings and outcomes are specific to the context, geographical area, and time frame defined in the scope, and may not be directly generalizable to other regions, non-logic-heavy subjects, or time periods without further validation.
- External factors such as network connectivity, hardware constraints, and user-specific behaviors are beyond the study's control and may influence real-world performance, particularly for latency-sensitive real-time mechanics such as live chat and synchronized scoring.

Tools to be used

9. Web Development Framework
    a. Next.js (React-based framework) — primary web development tool
10. Backend and AI
    a. Python v3.x — backend AI microservices
    b. LangChain — structuring the RAG pipeline and LLM capabilities
    c. OpenAI GPT-4o API — question generation and the vision-capable Solution Analyzer
    d. sentence-transformers/all-MiniLM-L6-v2 — embedding model
11. Database and Data Management
    a. PostgreSQL — local data storage for the web application
    b. PGVector — vector similarity search extension
    c. Prisma ORM — managing PostgreSQL queries
12. Real-Time Communication
    a. Socket.IO and Redis — low-latency real-time state synchronization of the multiplayer environment
13. Styling
    a. Tailwind CSS — design of the web application's user interface and responsiveness
14. Design and Prototyping
    a. Figma — high-fidelity prototypes for the professor dashboard and student quiz interfaces
15. Version Control and Deployment
    a. Git and GitHub — version control and repository management
    b. GitHub Actions — CI/CD pipeline for build, test, and deployment
    c. Docker — containerization of the Python FastAPI AI microservices
    d. Vercel — hosting platform for the Next.js web application
16. Testing Tools
    a. pytest — unit testing for Python microservices
    b. Jest — unit testing for the Next.js frontend
    c. Apache JMeter — performance, load, and stress testing

III. Project Milestones

The milestones below decompose QuizArena's scope into modules, features, and UI-level sub-components, sequenced from foundational infrastructure toward advanced, dependent functionality, and detailed enough to serve as a page-by-page development roadmap. The ordering reflects the project's dependency structure: authentication and the question bank must exist before content can be generated or validated; the Adaptive Randomization service must be available before the multiplayer engine can assign balanced teams; and the RAG pipeline, Solution Analyzer, and Professor Dashboard are layered on top of that working core. Two novelties are integrated directly into their relevant modules: Source Citation & Traceability within the RAG Question Generation Module (Milestone 5), and Misconception Analysis within the Solution Analyzer Module (Milestone 6), with a corresponding class-level Misconception Insights view surfaced in the Professor Dashboard (Milestone 7). Sub-components describe user-facing UI elements only (buttons, forms, tables, charts, modals, and similar controls); target dates are not specified as they are not fixed in the current project plan.

Milestone 1: Core Infrastructure & Authentication Module

1. **Feature: Sign Up / Login (Professor & Student)**
   a. Sign Up form (Name, UMak Gmail, Password, Role selection)
   b. Login form (Email, Password)
   c. Show/Hide password toggle
   d. "Forgot Password" link and reset-password form
   e. Inline field validation messages (invalid email, weak password, etc.)
   f. Loading indicator on Sign Up / Login buttons
   g. Role-based landing redirect (Professor Dashboard vs. Student Home)
2. **Feature: Section / Class Management**
   a. Add Class/Section button
   b. Edit Class button
   c. Archive Class button
   d. Section list table (Section Name, Subject, Professor, Student Count)
   e. Search bar for sections
   f. Filter by subject and semester
   g. Sort sections (name, student count, date created)
   h. Join Request notification badge
   i. Approve / Reject Join Request buttons
   j. Student roster table per section with Remove and Reassign actions
   k. Pagination

Milestone 2: Question Bank Module

1. **Feature: Manage Questions**
   a. Add Question button
   b. Edit Question button
   c. Delete / Archive Question button
   d. Search bar
   e. Filter by Subject, Difficulty, Topic, and Question Type
   f. Sort questions
   g. Question table
   h. Bulk import / export controls
   i. Pagination
   j. Preview Question modal
2. **Feature: Question Review & Approval**
   a. Pending Review tab
   b. Draft / Published status badge
   c. Approve button
   d. Reject button with reason input field

e. Edit-before-approving inline editor
f. Review history table (Date, Editor, Action)
g. Filter review queue by subject and submitted-by

Milestone 3: Adaptive Randomization Matchmaking Module

1. **Feature: Matchmaking Configuration**
    a. Enable / Disable Adaptive Randomization toggle (per battle session)
    b. Team size input field
    c. Fairness tolerance slider
    d. Preview Team Assignments button
    e. Team assignment preview table (Student, Team, Performance Level)
    f. Reshuffle Teams button
    g. Confirm Teams button
2. **Feature: Matchmaking Fairness Report**
    a. Fairness score badge (Balanced / Needs Review)
    b. Team balance chart per session
    c. Filter report by section and quiz session
    d. Export Fairness Report button

Milestone 4: Real-Time Multiplayer Quiz Battle Module

1. **Feature: Battle Room Lobby**
    a. Create Room button
    b. Join Room panel (Room Code input field)
    c. Mode selection control (Individual / Team / Battle Royale)
    d. Player list with avatars/status indicators
    e. Start Battle button (Professor only)
    f. Countdown timer display
    g. Leave Room button
2. **Feature: Live Battle Interface**
    a. Question display panel
    b. Answer input/selection area
    c. Live scoreboard sidebar
    d. Per-question timer / progress bar
    e. Speed Mode indicator badge
    f. Discussion Mode chat and voting panel
    g. Designated Leader badge and Confirm Final Answer button (leader only)
    h. Quick-reaction buttons
    i. Reconnecting/connection-status notification
3. **Feature: Battle Results**
    a. Final leaderboard table
    b. Individual / team score breakdown cards
    c. Play Again button
    d. Exit to Dashboard button
    e. Share Results button

Milestone 5: RAG Question Generation Module

1. **Feature: Syllabus Upload**
    a. Upload Syllabus/Document button
    b. Uploaded documents table (Filename, Upload Date, Status)
    c. Remove Document button
    d. Processing status badge (Uploading / Processing / Ready)
2. **Feature: AI Question Generator**
    a. Generate Question button
    b. Regenerate button
    c. Edit Generated Question button
    d. Approve Question button
    e. Reject Question button
    f. Search generated questions
    g. Filter by subject and topic
    h. Generated-question card with status badge (Pending / Approved / Rejected)
3. **Feature: Source Citation & Traceability**
    a. Retrieved source citation panel on every generated question
    b. Source document, topic/section, and page display
    c. Expand / Collapse source excerpt preview
    d. Source confidence badge (e.g., “Strong Match” / “Low Confidence Match”)
    e. “Flag Incorrect Citation” button
    f. Citation audit table listing generated questions with their linked source document and page
    g. Search/filter citation audit table by document or topic

Milestone 6: LLM-Driven Solution Analyzer Module

1. **Feature: Submission Handling**
    a. Upload handwritten math image button
    b. Programming code submission text/code input area
    c. Submission preview thumbnail
    d. Resubmit button
    e. Submission status badge (Pending / Analyzed)
2. **Feature: Solution Grading View**
    a. Analyze Submission button
    b. Step-by-step breakdown panel with per-step correct/incorrect indicators
    c. Per-step feedback comment bubbles
    d. Annotated image/code viewer highlighting the error location
    e. Overall grade badge
3. **Feature: Misconception Analysis**
    a. Analyze Submission button
    b. Student answer panel
    c. AI-generated misconception summary
    d. Misconception category badge (Sign Error, Off-by-One, Syntax Error, Logical Error, Computational Mistake, etc.)
    e. Common misconceptions chart (class-level)
    f. Student misconception history table
    g. Filter by class, topic, quiz, and misconception type
    h. Export Analytics button

Milestone 7: Professor Dashboard & Analytics Module

1. **Feature: Real-Time Match Analytics**
    a. Live match progress dashboard
    b. Live score cards per team/individual
    c. Match session selector dropdown
2. **Feature: Sortable Results Views**
    a. Results table with Sort-by-Accuracy control
    b. Sort-Alphabetically control
    c. Search bar for student/team names
3. **Feature: Misconception Insights**
    a. Class-level misconception overview chart
    b. Misconception trend indicator (increasing/decreasing icon)
    c. Filter by section, topic, and time period
    d. Drill-down link to a student's Misconception History (Milestone 6)
    e. Export Misconception Report button
4. **Feature: Engagement & Usage Reporting**
    a. Engagement summary cards (participation rate, average session length)
    b. Session/engagement log table
    c. Export Report button
5. **Feature: Session Replay**
    a. Match history list
    b. Replay playback controls (play, pause, step through questions)
    c. Replay summary panel

Milestone 8: System Integration, Testing & Deployment

1. **Functional & Usability Testing**
    a. Test cases derived from user-story acceptance criteria
    b. Validation of AI question generation accuracy and source citation correctness
    c. Validation of Solution Analyzer grading and misconception tagging accuracy
    d. Validation of professor editing and matchmaking configuration tools
2. **System-Wide Integration Testing**
    a. Cross-module workflow testing (question generation → battle → grading → dashboard)
    b. Real-time multiplayer synchronization testing across concurrent sessions
3. **Non-Functional Testing**
    a. Performance and load testing under classroom-scale concurrent use
    b. Stress testing under peak concurrent load
    c. Security testing of login, roles, and access permissions
4. **User Acceptance Testing & ISO/IEC 25010 Evaluation**
    a. UAT sessions with University of Makati faculty and students
    b. Likert-scale survey on ISO/IEC 25010 quality characteristics
5. **Deployment & Handover**
    a. Production deployment of the web application and AI services
    b. Post-deployment smoke testing across all modules
    c. End-user documentation (installation guide, professor manual, student quick-start guide)

IV. Project Organizational Structure

Stakeholders identified in the research comprise faculty members, students, and the development team within the College of Computing and Information Sciences at the University of Makati. The research identifies the following individuals as the project's development team; individual role assignments and specific responsibilities per member are not specified in the research.

Number	Name	Role
1	Canlas, Emerson B.	*Backend Developer*
2	Miguel, John D.	*Frontend Developer*
3	Picao, Mark Kevin N.	*System Analyst*
4	Villarica, Amrafel Marcus B.	*UI/UX Designer*

Additionally, the research notes that the development team is supported by a broader stakeholder group consisting of University of Makati faculty (who provide syllabi, validate AI-generated questions, and participate in User Acceptance Testing) and students (who participate in multiplayer quiz battles and UAT feedback sessions).